Article published online: 2024-10-22

Chronotype, sleep hygiene, and sleep characteristics among athletes of different disciplines

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Keywords
Sleep, Athlete, Chronotype, Sport, sexes

accepted 16.09.2024
published online 2024

Bibliography
Int J Sports Med
DOI 10.1055/a-2421-9310
ISSN 0172-4622
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Georg Thieme Verlag KG, Rüdigerstraße 14,
70469 Stuttgart, Germany

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ABSTRACT

The objective of the study was to assess chronotype, sleep hygiene, and sleep characteristics of Italian athletes in order to identify differences among sport disciplines (team- vs individual-sports), level of competition (Elite vs Non-Elite) and sexes. Moreover, correlations between chronotype, sleep hygiene and sleep quality were evaluated. Two hundred eighty Italian athletes (Females:113; Males:167) participated in the study. They completed the Pittsburgh Sleep Quality Index (PSQI), the Sleep Hygiene Index (SHI) and the MorningnessEveningness Questionnaire (MEQ). Among the athletes, 66.4 % were intermediate-type, 21.8 % were morning-type and 11.8 % were evening-type. 55.4 % of the sample was classified as good sleeper (PSQI < 5), while 16.1 % had poor sleep hygiene (SHI ≥ 35). Females had higher total sleep time ($p = 0.022$) than males. Individual-sports showed more morning-oriented score ($p = 0.001$) and lower PSQI ($p = 0.006$) and SHI ($p < 0.001$) scores, with earlier timing of sleep (bedtime:$p < 0.001$; wake-up time:$p < 0.001$) compared to team-sports. Elite athletes exhibited greater sleep efficiency ($p = 0.009$), and lower sleep latency ($p = 0.018$) and wake after sleep onset ($p = 0.043$) than Non-Elite athletes. Additionally, significant correlations were found between PSQI and SHI ($r = 0.43$, $p < 0.001$), and between MEQ and SHI ($r = -0.35$, $p < 0.001$). This study yields crucial insights into the sleep behaviors of Italian athletes, revealing that nearly half of the sample were bad sleepers.

Introduction

Sleep is a fundamental aspect of human physiology and plays a crucial role in maintaining overall health and well-being [1]. Considering the high variability in sleep among athletes [2], it becomes imperative to examine various endogenous and exogenous factors. These include, but are not limited to, chronotype, sport disciplines, level of competition, sexes, and baseline sleep hygiene. Athletes can perform better if they align their training sessions and competitions with the times of the day when they are naturally most awake and alert. In a study of Lastella et al. [3], morning-type athletes were more likely to select and participate in sports requiring morning training, whereas evening-type athletes selected and participated in sports requiring training in the evening. Based on this, recent studies suggest that an athlete's sleep quantity and sleep schedule may be influenced by the sport in which they are involved [4]. This can be attributed to numerous factors, including the distinct demands of the sport, variations in training schedules, and the timing of competitions. If not managed optimally, training, competitions, and mood

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factors can lead to a reduction in sleep hygiene. Sleep hygiene is defined as a set of practices and habits that promote healthy and consistent sleep patterns. It involves adopting behaviors and creating an environment that optimizes the quality and quantity of sleep [5]. In terms of sleep characteristics, the National Sleep Foundation asserts that adults need 7–9 hours of sleep per night to maintain optimal health and functioning [6]. However, athletes often sleep less than 7 hours [7] and this can lead to negative effects in physical and cognitive functioning, mood disturbances, decreased alertness, and heightened daytime sleepiness episodes [8–10]. It is not just the type of sport that can impact on the sleep hygiene of athletes; literature showed that there are sex differences in sleep characteristics [11]. In a study by Silva et al., which assessed 146 athletes from the Brazilian Olympic Team, it was demonstrated that females exhibited better sleep quality than males (sleep efficiency: +6.27 %; sleep latency: −15.5 minutes), although no difference was detected in total sleep time. Additionally, 53 % of the total sample reported some sleep complaints, with a higher prevalence among male athletes (61 %) compared to female athletes (39 %) [12]. Also, another factor that could affect athletes' sleep is the level of competition. Sleep differences are often associated with the time spent on sport practice. Non-elite athletes, for instance, may not train as extensively as elite athletes, as they need to allocate more time to school and work. Consequently, provincial and regional-level athletes tend to have more non-sporting activities than national and international-level athletes, potentially impacting athletes' schedules, sleep quality, quantity, and sleep/wake regularity [13].

For these reasons, the primary aim of the present study was to examine and study the chronotype, sleep hygiene, and sleep characteristics of athletes enrolled in Italian sports federations. The goal was to identify differences between sport disciplines (team vs. individual sports), levels of competition (elite vs. non-elite), and sexes. In addition, to better comprehend the relationship between sleep, chronotype, and sleep hygiene, the second aim of the study was to assess correlations among these variables in the athlete population. Our initial hypotheses were to identify significant differences in sleep characteristics and chronotype across sexes, sport disciplines and level of competition. In detail, we expected that females had better sleep quality than males, individual sports athletes had more morning-oriented score than team sports athletes, with an anticipate timing of sleep. Moreover, we expected that elite athletes had a better sleep hygiene and a greater total sleep time than Non-elite athletes. Regarding correlations, our expectation was to find significant association between sleep quality and sleep hygiene. Regarding the relationship between sleep hygiene and chronotype, which has been minimally studied in the literature, we hypothesized no significant correlations between these variables.

Materials and Methods

Study design

This was a cross-sectional observational study in accordance with the STROBE guidelines [14]. The study was approved by the Ethical Committee of < institution affiliated with one of the authors > (ref. no.: < number of Project >), and all procedures were performed in compliance with laws and regulations governing the use of human subjects (Declaration of Helsinki). All participants received an explanation of the purpose of the study and its methods, and a written informed consent was obtained from all subjects. Approximately 20 minutes were required to complete the survey and the results were reported following specific guidelines [15].

Power analysis

Based on existing study that compared sleep between team and individual sports [7], the effect size (d) was estimated from the total sleep time results. To perform the power analysis, an α level with $p = 0.05$, a power of 0.85 and an effect size (d) of 0.434 was selected. It was estimated that 97 subjects would be needed for each group. The sample size calculation was performed with the use of the G-Power software (G-Power version 3.1, Düsseldorf, Germany).

Participants

During the competitive period of 2022/2023 season, athletes registered with Italian sport federations were enrolled for this study. Participants were screened for eligibility using inclusion criteria: male or female, age between 18 and 40 years, Italian athletes, training for three or more times per week and hours of training/week over 2.5. The exclusion criteria were: any medical condition influencing sleep and the frequency of training for the last month, training or sleeping in altitude, had undertaken a transmeridian travel in the 4 weeks preceding the study, or any medication for sleep disorders. Participants who met the above criteria were recruited in the study and written informed consent was obtained from all participants. Athletes were recruited from sports club centers upon invitation by authors or coaches, and they received by e-mail the online survey in order to answer to all questions. Data were subsequently exported to an Excel file (Microsoft, Redmond, Washington, USA) for data analysis. The sample was subsequently divided based on three factors: sexes, level of competition, and sports discipline (▶ **Fig. 1**).

Regarding the level of competition, athletes competing at the provincial and regional levels were considered non-elite, whereas those competing at the national and international levels were classified as elite. Taking into account the discipline, the sample was divided into team sports and individual sports.

Survey

The survey was composed by five sections: i) anthropometric and demographic data, and training investigation (e. g. sexes, height and body mass, sport discipline, and trainings characteristics); ii) Morningness-Eveningness Questionnaire (MEQ); iii) Pittsburgh Sleep Quality Index (PSQI); iv) Sleep Hygiene Index (SHI); v) information regarding their ideal sleep, if they could choose when and how much to sleep.

Morningness-Eveningness Questionnaire (MEQ)

The MEQ is a tool used to assess participants' circadian rhythm or their natural preference for being either a "morning person" or an "evening person" [16]. The MEQ consists of a series of questions designed to determine a person's chronotype, which is their tendency to feel more alert and energetic at specific times of the day. The MEQ score ranges from a minimum of 16 to a maximum of 86 (16–30: "definite evening": 31–41: "moderate evening"; 42–58: "intermediate"; 59–69: "moderate morning"; 70–86: "definite morning").

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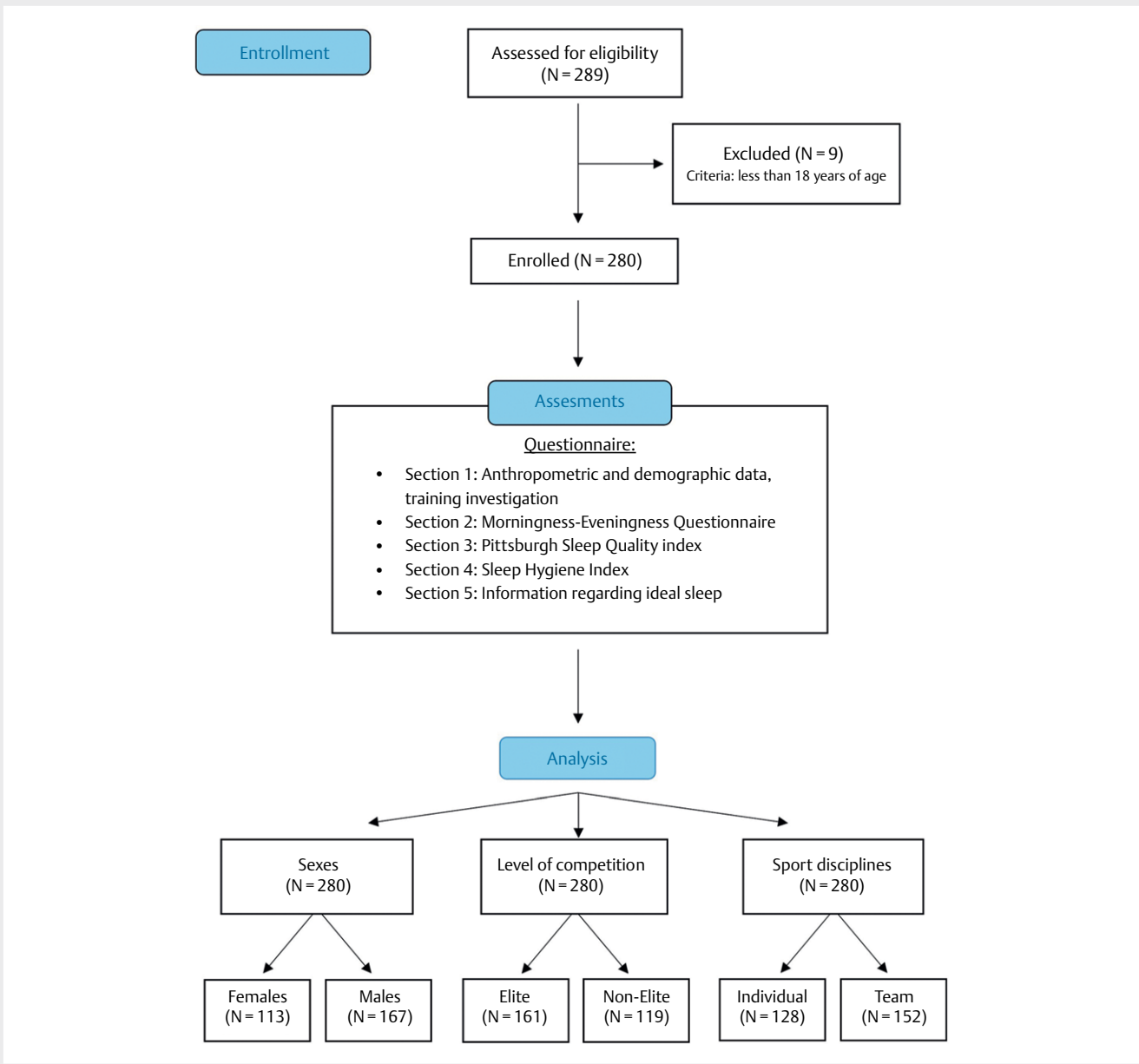

▶ **Fig. 1** Flowchart of study design.

Pittsburgh Sleep Quality Index (PSQI)

Sleep quality over the last month interval was assessed with the Italian version of the Pittsburgh Sleep Quality Index [17]. The PSQI is a widely used tool to assess an individual's sleep quality and identify potential sleep disturbances [18]. It consists of a series of questions that provide valuable insights into different aspects of sleep patterns and difficulties experienced over the past month. This questionnaire is composed of 19 questions addressing seven components: subjective sleep quality, sleep latency, sleep duration, sleep efficiency, sleep disturbances, sleep medication, and daytime dysfunction. Each component includes items with 4-point Likert scales (0–3). The sleep quality global score is the sum of all components, and the score ranges from 0 to 21. Scores higher than 5 indicate poor sleep quality. *For this study, the validated Italian version of PSQI was used* [17].

Sleep Hygiene Index (SHI)

The SHI is a self-report questionnaire developed to evaluate the practice of sleep hygiene behaviors with adequate psychometric properties in different samples [19–21]. The SHI questionnaire consists of 13 items that investigate how often a subject performs specific behaviors. The answers have a score ranging from 1 to 5 (1 = "never", 2 = "rarely", 3 = "sometimes", 4 = "frequently", 5 = "always") and the total score is obtained by adding the score of each individual item and ranges from 13 to 65. Higher scores indicate poor sleep hygiene; previous study used a score of ≤ 26 as good sleep hygiene habits, 27–34 as average, and ≥ 35 as poor sleep

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hygiene [19]. *For this study, the validated Italian version of SHI was used* [19].

Sleep Variables

Subjective sleep parameters (wake-up time, bedtime, sleep latency, total sleep time) was obtained from questions 1, 2, 3, 4 of PSQI. Wake After Sleep Onset (WASO) was obtained by a question of section 5 of the survey: "During the past month, how many minutes on average you were awake during the night?" Sleep efficiency was calculated with the following formula in order to obtain more specific data:

Sleep efficiency = [(Time in bed – Sleep latency – Wake After Sleep Onset) / Time in bed] x 100

Statistical Analysis

Data from questionnaires and sleep variables were checked with the Shapiro-Wilk test. In case of non-normal distribution of data (age, body mass, Body Mass Index, number of training sessions per week, time of training per week, SHI, PSQI, total sleep time, sleep efficiency, sleep latency, WASO, bedtime, wake-up time, and all SHI items), a Mann-Whitney test was performed to assess differences between sexes, level of competition, and sport disciplines; in addition, an unpaired t-test was performed for MEQ and height (normal distribution of data). Given a non-normal distribution, all correlations were performed with using a Spearman test, and were considered significant with $r > 0.3$ and $p < 0.05$ [22]. Effect sizes for pairwise comparison were calculated using Cohen's d and considered to be either trivial (effect size: < 0.20), small (0.21–0.60), moderate (0.61–1.20), large (1.21–2.00), or very large (> 2.00) [23]. The level of significance was set at $p < 0.05$. Statistical analysis was performed using GraphPad Prism (GraphPad Software, San Diego, CA).

Results

Characteristics of participants

A total of 280 athletes (females: 113; males: 167) registered with Italian sport federations completed the survey during competitive period of 2022/2023 season (from November 2022 to June 2023). Regarding sport disciplines, 152 athletes practiced team sports (soccer, volleyball, basketball, baseball, flag football, handball) and 128 individual sports (track and field, judo, fencing, swimming, rowing, snowboarding). Based on their level of competition, athletes were divided in two categories: elite (n = 161, competing at international and national level) and non-elite (n = 119, competing at regional and provincial level). ▶ **Table 1** shows the anthropometric data and characteristics of the sample.

MEQ, PSQI and SHI

▶ **Fig. 2** shows violin plots for MEQ, SHI, PSQI scores (panel a, b, and c) and sleep variables (panel d – i), also expressed as mean ± SD and 95 % CI, for the entire study sample. In detail, the sample included: 33 evening-type (11.8 %), 186 intermediate-types (66.4 %), and 61 morning-types (21.8 %). Eighty-three athletes had good sleep hygiene (29.6 %), 152 athletes had medium sleep hygiene scores (54.3 %), and 45 athletes had poor sleep hygiene (16.1 %). In addition, 155 athletes (55.4 %) were classified as good sleepers, whereas 125 (44.6 %) as poor sleepers.

Training characteristics

Our survey examined the training characteristics of all athletes (e. g. number of training sessions per week, minutes of training per week, average minutes of training for each session, etc.). The results revealed interesting trends. On average, elite athletes trained more frequently (elite: 5.5 trainings/week; non-elite: 3.1 trainings/week; $p < 0.001$) and for a significantly higher time (elite: 701.7 minutes/week; non-elite: 358.4 minutes/week; $p < 0.001$) compared to their non-elite counterparts. Additionally, we observed that athletes in individual sports tend to train more frequently (individual: 6.0 trainings/week; team: 3.7 trainings/week; $p < 0.001$) and for longer time (individual: 662.1 minutes/week; team: 466.3 minutes/week; $p < 0.001$) compared to team sports athletes.

Sleep characteristics: differences between sexes, sport disciplines, and level of competition

▶ **Table 2** shows differences between sexes, sport disciplines (individual vs. team sports) and level of competition (elite vs. non-elite) for all variables considered. Females had a higher total sleep time (+ 14 minutes) than males; individual sports had a greater score in MEQ (+ 3.3) and an anticipated timing of sleep (bedtime: -36 minutes, wake-up time: −28 minutes) than team sports; elite athletes had greater sleep efficiency (+ 2.8 %), and lower sleep latency (-2.1 minutes) and WASO (−11.3 minutes) than non-elite athletes. ▶ **Table 3** shows differences between sexes, sport disciplines, and level of competition each single item of SHI. Females had a lower score in item 7 ($p = 0.004$) and a higher score in item 8 ($p = 0.006$) compared to males. Individual sport athletes had lower scores in item 1 ($p = 0.022$), item 2 ($p = 0.027$), item 4 ($p < 0.001$), item 5 ($p = 0.003$), item 7 ($p = 0.026$), and item 9 ($p = 0.029$) compared team sport athletes. Finally, elite athletes had lower scores in item 4 ($p = 0.002$), item 6 ($p = 0.002$), and a higher score in item 11 ($p = 0.002$) compared to non-elite athletes.

Correlations

▶ **Fig. 3** shows the correlation results between questionnaires' scores. Significant and positive correlation between PSQI and SHI ($r = 0.43$, $p < 0.001$) and significant and negative correlations between MEQ and SHI ($r = -0.35$, $p < 0.001$) were detected.

Discussion

The main findings of the present study include: i) females sleep more than males; ii) team sport athletes tend to be more evening-types, and they exhibit poorer sleep quality and sleep hygiene compared to individual sport athletes; iii) elite athletes demonstrate better sleep efficiency, reduced sleep latency, and wake less after sleep onset compared to non-elite athletes; iv) a positive correlation was found between PSQI and SHI, while a negative correlation between MEQ and SHI was detected.

Athletic success hinges on a combination of various factors, encompassing rigorous physical and mental training, optimal nutrition, the preservation of muscle and bone health, and the capacity to prioritize adequate recovery and rest between training sessions

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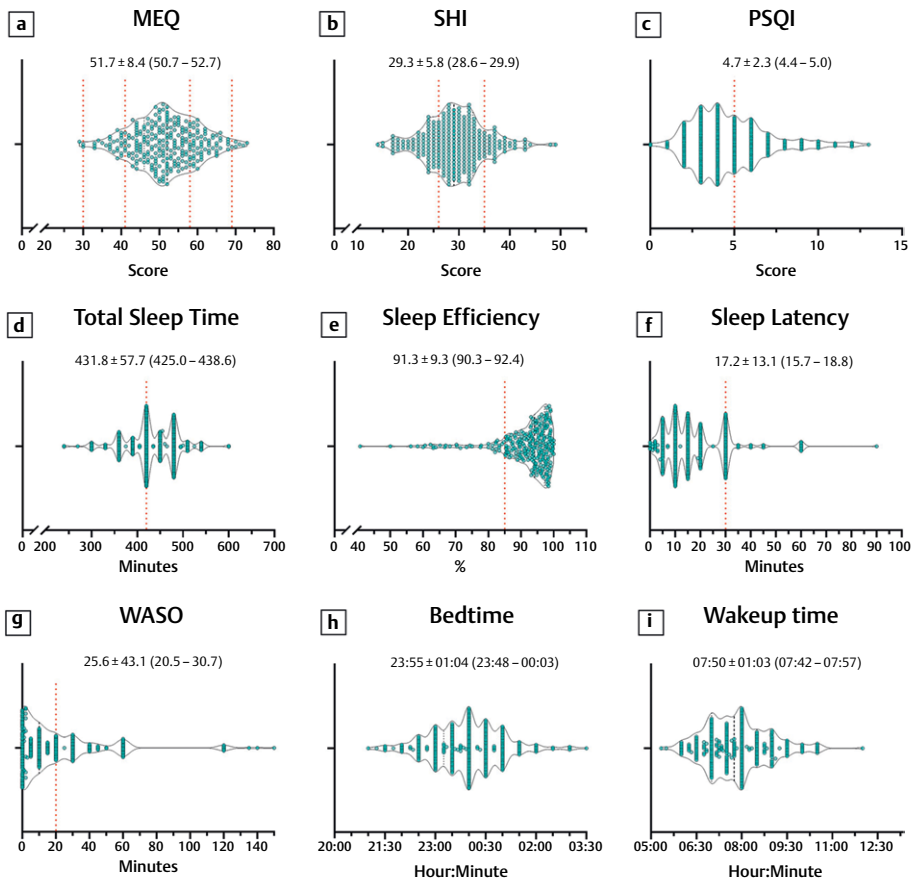
▶ **Table 1** Anthropometric and demographic characteristics of sample.

	Total (N: 280)	Females (N: 113)	Males (N: 167)	Team (N: 152)	Individual (N: 128)	Elite (N: 161)	Non-Elite (N:119)
Age (years)	24.7 ± 6.2	25.5 ± 7.5	24.6 ± 6.2	24.7 ± 5.5	24.3 ± 6.3	23.0 ± 4.6	**26.5 ± 6.6*** **	
Body mass (kg)	70.9 ± 13.0	61.9 ± 9.4	**77.0 ± 11.6*** **	74.7 ± 13.1	**66.4 ± 11.4*** **	71.3 ± 12.7	70.4 ± 13.5
Height (cm)	176.2 ± 9.2	169.9 ± 9.4	**180.5 ± 7.5*** **	177.9 ± 9.2	**174.2 ± 8.8*** **	177.1 ± 9.3	**174.9 ± 9.0***
BMI	22.7 ± 3.0	21.4 ± 2.5	**23.6 ± 3.0*** **	23.5 ± 3.0	**21.8 ± 2.8*** **	22.6 ± 2.8	22.9 ± 3.3

Legend. Data of age, body mass, height, and BMI are reported as mean ± SD. BMI: Body Mass Index; *: p < 0.05; **: p < 0.01; ***: p < 0.001.



▶ **Fig. 2** Questionnaire score and sleep variables of sample. ***Legend.*** *Data are reported as mean ± SD (95 % CI). Red lines show validated cut-off values of questionnaires and sleep variables. Panel a: MEQ score; Panel b: SHI score; Panel c: PSQI score; Panel d: Total Sleep Time; Panel e: Sleep Efficiency; Panel f: Sleep Latency; Panel g: WASO; Panel h; Bedtime; Panel i: Wakeup Time. MEQ: Morningness-Eveningness Questionnaire; SHI: Sleep Hygiene Index; PSQI: Pittsburgh Sleep Quality Index; WASO: Wake After Sleep Onset.*

and competitions [24]. When studying the impact of recovery on athletic performance, it is essential to consider the athlete's chronotype and their habitual sleep schedule [25]. Analyzing our data from MEQ, we found that 66.4 % of athletes were intermediate-type, 21.8 % were morning-types and 11.8 % were evening-types. It has been observed that the chronotype presents differences in circadian rhythms for several variables (e. g. perception of effort, personality, and cognitive function) that can influence performance [26]. In general, morning-types are used to waking up and going to bed earlier, and their peak performance occurring in the morning. Conversely, evening-types go to bed and wake up later, with their peak performance in the evening [27]. The results of our study align partially with the existing literature. In fact, it has been demonstrated that 60 % of the adult population is classified as intermediate type, with the remaining 40 % falling into the other categories [28, 29]. Based on PSQI, SHI, and sleep variables, the results revealed that 55.4 % of our sample was classified as a good-sleeper (PSQI < 5), 16.1 % had poor sleep hygiene (SHI ≥ 35), 24.6 % didn't sleep enough during night (< 7 h), 24.6 % presented high sleep latency (> 30 minutes), and 13.6 % had an insufficient sleep efficiency

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▶ **Table 2** Differences between sexes, sport disciplines, and level competition.

	MEQ	SHI	PSQI	TST	SE	SL	WASO	Bedtime	Wake-up time
Females	52.0 ± 8.4 (50.4–53.5)	29.5 ± 5.8 (28.5–30.6)	4.4 ± 5.2 (4.4–5.2)	440.4 ± 57.4 (429.7–451.0)	90.6 ± 11.8 (88.4–92.8)	19.3 ± 21.4 (15.3–23.3)	28.0 ± 49.6 (18.7–37.2)	23:49 ± 01:07 (23:36–00:01)	07:54 ± 01:06 (07:41–08:06)
Males	51.5 ± 8.5 (50.2–52.8)	29.1 ± 5.9 (28.1–30.0)	4.6 ± 2.4 (4.2–4.9)	**426.0 ± 57.3*** (417.3–434.8)	91.1 ± 9.6 (89.6–92.6)	16.8 ± 11.5 (15.0–18.5)	24.0 ± 38.2 (18.2–29.9)	00:04 ± 01:17 (23:52–00:15)	07:46 ± 01:05 (07:36–07:56)
ES	–	–	–	0.25, small	–	–	–	–	–
Individual Sports	53.5 ± 8.9 (51.9–55.0)	27.8 ± 5.8 (26.8–28.8)	4.3 ± 2.4 (3.9–4.8)	437.0 ± 52.9 (427.7–447.2)	92.2 ± 7.7 (90.8–93.5)	17.0 ± 14.6 (14.5–19.6)	21.2 ± 30.0 (15.9–26.4)	23:38 ± 01:06 (23:27–23:50)	07:34 ± 01:02 (07:03–07:45)
Team Sports	**50.2 ± 7.7**** (49.0–51.4)	**30.5 ± 5.6***** (29.6–31.4)	**5.0 ± 2.3**** (4.6–5.3)	427.4 ± 61.2 (417.6–437.2)	89.9 ± 12.3 (87.9–91.9)	18.5 ± 17.6 (15.6–21.3)	29.6 ± 51.5 (21.2–37.6)	**00:14 ± 01:15***** (00:02–00:26)	**08:02 ± 01:05***** (07:51–08:12)
ES	0.40, small	0.47, small	0.30, small	–	–	–	–	0.51, small	0.44, small
Elite	52.0 ± 8.3 (50.7–53.3)	29.4 ± 5.9 (28.5–30.3)	4.5 ± 2.4 (4.2–4.9)	433.6 ± 58.2 (424.5–442.7)	92.1 ± 9.2 (90.7–93.5)	16.9 ± 17.8 (14.1–19.7)	20.8 ± 33.4 (15.6–26.0)	23:59 ± 01:20 (23:47–00:12)	07:47 ± 01:08 (07:36–07:58)
Non-Elite	51.3 ± 8.6 (49.7–52.9)	29.1 ± 5.8 (28.0–30.1)	4.9 ± 2.2 (4.5–5.3)	429.4 ± 57.1 (419.0–439.7)	**89.3 ± 12.0**** (87.2–91.5)	**19.0 ± 13.9*** (16.5–21.6)	**32.1 ± 53*** (22.5–41.8)	23:55 ± 01:02 (23:44–00:07)	07:52 ± 01:02 (07:41–08:03)
ES	–	–	–	–	0.26, small	0.13, trivial	0.26, small	–	–

Legend. Data are reported as mean ± SD (lower and upper 95 %CI). MEQ: Morningness-Eveningness Questionnaire; SHI: Sleep Hygiene Index; PSQI: Pittsburgh Sleep Quality Index; TST: Total Sleep Time; SE: Sleep Efficiency; SL: Sleep Latency; WASO: Wake After Sleep Onset; ES: Effect Size; *: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$.

(< 85 %). Previous studies examined sleep characteristics of athletes. Mah et al. [30] assessed the sleep of 628 collegiate student-athletes at Stanford University. They found that 42.4 % of athletes were identified as poor sleepers with PSQI, and 39.1 % reported a mean total sleep time < 7 hours. Carter et al. [31] monitored 121 collegiate athletes with validated questionnaires and actigraphy for three consecutive nights, revealing that 35 % of them were classified as a poor-sleepers and 61 % as a short-sleepers. In terms of sleep hygiene, higher values were registered for items 2, 3, 7, 8, 9, and 13. These items investigated the regularity of sleep timing (bedtime and wake-up time), emotional status (e. g. stress, angry, worry, etc.), and active behaviors before sleep. Regarding sleep timing, the body's natural clock has a profound effect on most biological functions, making it an important factor in optimizing sleep duration [32]. Poor sleep can result when individuals attempt to sleep outside of their circadian phase, and this may also impact the peak performance of athlete [33]. For these reasons, athletes should maintain a consistent bedtime and wake-up time throughout the week. Regarding the emotional status, studies indicate a high prevalence of sleep complaints in individuals with anxiety [34, 35]. In particular, insomnia affects approximately 50 % of individuals with anxiety, and insufficient sleep could instigate or further exacerbate it [35, 36]. Finally, active sleep behaviors before bedtime (e. g. studying, performing high-intensity exercise, working or using electronic devices) are recognized as bad habits capable to compromise the quality of sleep [33, 37]. For most people, engaging in moderate-intensity exercise typically does not adversely affect their sleep, especially if the exercise session concludes at least 90 minutes before bedtime [38]. This interval allows sufficient time for endorphin levels and core body temperature to return to levels that are conducive to sleep. In preparation for sleep, body temperature drops, heart rate slows, and brain waves become slower [39]. Conversely, exercise and active behaviors can lead to an elevation in core body temperature, heart rate, and brain activation, all of which can potentially disrupt sleep. Due to these underlying physiological processes, experts generally discourage engaging in such activities just before bedtime [40].

It is well known that different sports require distinct training and engagements. Also, non-elite athletes often have different daily routines compared to elite athletes. For these reasons, we analyzed difference between sport disciplines (team vs. individual sports), levels of competition (elite vs. non-elite), and sexes. The choice to divide the sample based on the level of competition (Non-elite: provincial and regional level; elite: national and international level) is driven by the fact that higher-level athletes correspond to more intensive training programs to reach the success. This is confirmed by the results obtained regarding the difference in the numbers of weekly training sessions (elite: 5.5 trainings/week; non-elite: 3.1 trainings/week; $p < 0.001$) and the total time spent on weekly training (elite: 701.7 minutes/week; non-elite: 358.4 minutes/week; $p < 0.001$). Analyzing the differences between team and individual sports athletes, a statistically significant result emerged in MEQ scores ($p = 0.001$). Specifically, team sports athletes tend to be more evening-type compared to individual sports athletes (team: 50.2 ± 7.7; individual: 53.5 ± 8.9). Indeed, individual athletes had an earlier bedtime (36 minutes earlier) and wake-up time (26 minutes earlier) compared to team athletes ($p < 0.001$). In terms of PSQI and SHI, team athletes

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▶ **Table 3** Differences between sexes, sport disciplines, and level of competition for all items of SHI.

	Females	Males	ES	Individual	Team	ES	Elite	Non-Elite	ES
1. I take daytime naps lasting two or more hours.	1.69 ± 0.96	1.75 ± 1.00	0.554	1.62 ± 0.99	**1.82 ± 0.97** *	**0.20, small**	1.81 ± 1.05	1.62 ± 0.87	0.209
2. I go to bed at different times from day to day.	2.66 ± 0.80	2.56 ± 0.99	0.261	2.50 ± 0.87	**2.71 ± 0.95** *	**0.23, small**	2.60 ± 0.85	2.60 ± 1.00	0.803
3. I get out of bed at different time from day to day.	2.39 ± 0.88	2.29 ± 0.93	0.327	2.27 ± 0.93	2.40 ± 0.90	0.217	2.35 ± 0.89	2.31 ± 0.95	0.662
4. I exercise to the point of sweating within 1 hour of going to bed.	1.55 ± 0.88	1.55 ± 0.95	0.607	1.27 ± 0.62	**1.78 ± 1.06** ***	**0.59, small**	1.40 ± 0.79	**1.76 ± 1.04** **	**0.39, small**
5. I stay in bed longer than I should two or three times a week.	2.38 ± 1.14	2.16 ± 1.11	0.094	2.07 ± 1.18	**2.40 ± 1.06** **	**0.29, small**	2.23 ± 1.14	2.28 ± 1.11	0.657
6. I use alcohol, tobacco, or caffeine within 4 hours of going to bed or after going to bed.	2.06 ± 1.26	2.07 ± 1.26	0.999	1.90 ± 1.15	2.20 ± 1.33	0.082	1.87 ± 1.16	**2.33 ± 1.34** **	**0.37, small**
7. I do something that may wake me up before bedtime (for example: play video games, use the internet, or clean).	2.61 ± 1.14	**3.06 ± 1.33** **	**0.36, small**	2.70 ± 1.31	**3.03 ± 1.22** *	**0.26, small**	2.98 ± 1.30	2.74 ± 1.22	0.107
8. I go to bed feeling stressed, angry, upset, or nervous.	2.55 ± 0.74	**2.32 ± 0.85** **	**0.29, small**	2.40 ± 0.77	2.42 ± 0.86	0.860	2.40 ± 0.81	2.42 ± 0.83	0.908
9. I use my bed for things other than sleeping or sex (for example: watch television, read, eat, or study).	2.85 ± 1.29	2.80 ± 1.36	0.765	2.63 ± 1.35	**2.98 ± 1.29** *	**0.27, small**	2.89 ± 1.38	2.71 ± 1.25	0.267
10. I sleep on an uncomfortable bed (for example: poor mattress or pillow, too much or not enough blankets).	1.69 ± 1.05	1.63 ± 0.94	0.989	1.61 ± 1.03	1.70 ± 0.96	0.171	1.74 ± 1.03	1.54 ± 0.94	0.064
11. I sleep in an uncomfortable bedroom (for example: too bright, too stuffy, too hot, too cold, or too noisy)	1.60 ± 0.90	1.63 ± 0.90	0.769	1.52 ± 0.83	1.70 ± 0.94	0.076	1.75 ± 0.94	**1.44 ± 0.80** **	**0.36, small**
12. I do important work before bedtime (for example: pay bills, schedule, or study).	2.40 ± 1.05	2.22 ± 1.03	0.228	2.35 ± 1.11	2.25 ± 0.99	0.508	2.40 ± 1.08	2.15 ± 0.97	0.074
13. I think, plan, or worry when I am in bed.	3.12 ± 0.99	3.00 ± 1.01	0.438	2.98 ± 1.07	3.11 ± 0.95	0.232	3.00 ± 1.02	3.16 ± 0.98	0.178

Legend. Data are reported as Mean ± SD. ES: Effect Size; *: $p<0.05$; **: $p<0.01$; ***: $p<0.001$.

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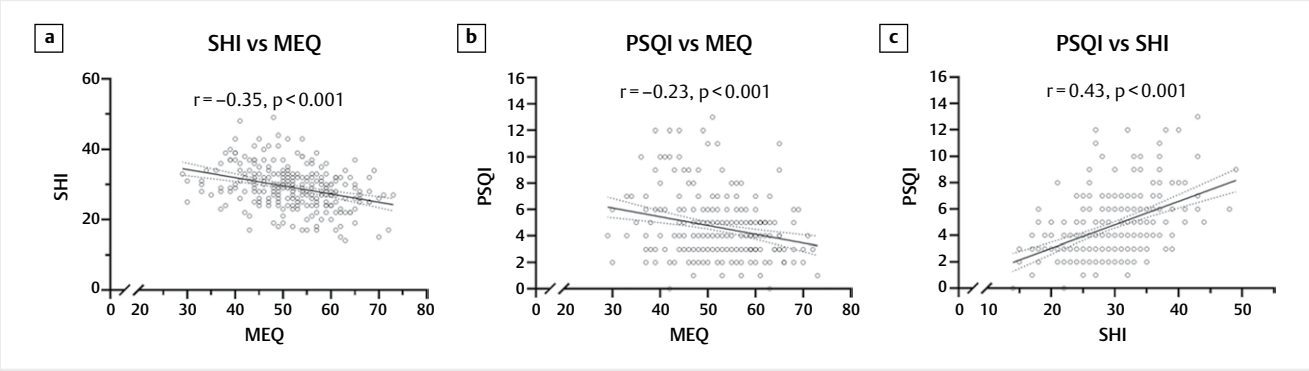

▶ **Fig. 3** Correlations between questionnaires' scores. ***Legend.*** *Panel a: Correlations between SHI and MEQ; Panel b: Correlation between PSQI and MEQ; Panel c: Correlation between PSQI and SHI; MEQ: Morningness-Eveningness Questionnaire; SHI: Sleep Hygiene Index; PSQI: Pittsburgh Sleep Quality Index; r: Spearman's coefficient; p: pvalue.*

exhibited poorer sleep compared to individual athletes, showing lower sleep quality ($p = 0.006$) and poorer sleep hygiene ($p < 0.001$). Despite the poorer values observed in team sports, it is interesting to note that there were no significant differences in sleep variables between groups. Our results contrast with findings from other studies. In the study by Lastella et al. [7], sleep differences among athletes from team and individual sports were evaluated. They found that individual athletes slept less (−30 minutes), had lower sleep efficiency (−0.5 %), and experienced longer sleep latency (+6 minutes) compared to team athletes. However, a study of Driller et al. [41] assessed the sleep of 407 elite adult athletes across sixteen sports, including both individual and team sports. Results revealed that individual athletes had higher total sleep time (+17 minutes) and sleep efficiency (+2.5 %). The differences in training characteristics and schedule between sport disciplines may influence the amount and quality of sleep. These training demands could also be influenced by the competitive level. Specifically, national and international level athletes showed higher sleep efficiency (+2.8 %, $p = 0.009$), lower sleep latency (−2.1 minutes, $p = 0.018$), and lower WASO (−11.3 minutes, $p = 0.043$) compared to regional and provincial level athletes. These results may be influenced by the fact that non-elite athletes cannot dedicate as much time to training as elite athletes, as they have to allocate more time to school and work [42]. Consequently, provincial and regional level athletes tend to engage in more non-sporting activities than national and international level athletes, potentially affecting athletes' schedules, sleep quality and quantity and sleep/wake regularity [13]. It has been studied that women have better sleep quality than men with a longer total sleep time, shorter WASO, better sleep efficiency, and shorter sleep latency [43]. However, from our results, there is no significant difference in sleep quality between sexes. Instead, it emerges that the total sleep time is significantly greater in women compared to men ($p = 0.022$; +14 minutes). This can be attributed to the fact that, on average, women have a bedtime that is 15 minutes earlier than men and a wake-up time that is 8 minutes later than men. These results contrast with the study by Silva et al. [12], where, analyzing the sleep of Brazilian athletes, women showed better sleep quality than men but not a longer total sleep time. It is important to note that, even though men and women have similar sleep requirements, the risk and nature of sleep disorders can differ between sexes [44]. Considering the type of sport, it is noteworthy that several disparities among them can influence the need for sleep. These variations include factors such as training volume and intensity, training schedules, the psychological stress associated with training, and the integration of training with other life commitments, such as work or education [45].

In terms of differences in SHI items, females had a lower score in item 7 (active behaviors before bedtime) and a higher score in item 8 (stress, angry or nervous before sleep) compared to males. Individual sport athletes had lower scores in item 1 (naps), item 2 (unregular bedtime during week), item 4 (exercise before bedtime), item 5 (stay in bed longer after wake-up), item 7, and item 9 (use of bed for activities other sleeping and sex) compared to team sport athletes. Finally, elite athletes had lower scores in item 4, item 6 (use of alcohol, tobacco, or caffeine), and higher score in item 11 (sleep in an uncomfortable bedroom) compared to non-elite athletes. These results are interesting because they provide more information regarding the behaviors and emotional status of sample. The prevalence of negative emotional status in females before bedtime was found in our sample, and it aligns with studies in the literature that confirm the predominance of anxiety, psychological distress, and chronic stress disorders in females compared to males [46, 47]. In terms of sport disciplines, team sports athletes exhibit higher scores on the SHI items. However, these differences do not appear to impact their sleep, as there are no significant differences in the quantitative and qualitative sleep parameters when compared to athletes participating in individual sports. On the contrary, the higher scores in SHI items assessed in non-elite athletes compare to elite ones could have a negative impact on the qualitative variables of their sleep (SE, SL, and WASO).

Regarding the second aim of the study, we found a positive correlation between PSQI and SHI scores ($r = 0.43$, $p < 0.001$), demonstrating that poor sleep quality is associated with poor sleep hygiene. Our results align with findings from other studies. Biggins et al. [48] investigated the prevalence of clinically relevant sleep problems in 58 elite international athletes and their associations with sleep hygiene and other conditions. Their results indicated

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that athletes who had relevant sleep problems were more likely to have poorer sleep hygiene practices ($p < 0.002$). This could be linked to numerous components and behaviors that have already been demonstrated to affect sleep, such as the consumption of caffeine [49], the use of smartphones and electronic devices [37], light exposure [50], or temperature [51]. In a recent study by Vitale et al. [5], it has been demonstrated that the adoption of sleep hygiene strategies (practices and habits with the aim to improve the quality and duration of sleep) could increase total sleep time and reduce sleep latency in elite athletes. Also, analysis did not detect a correlation between PSQI and MEQ, but a negative correlation between MEQ and SHI ($r = -0.35$, $p < 0.001$) was found, indicating that morning-types are associated with better sleep hygiene. This is an important result because many studies demonstrated the association between chronotype and sleep quality [52–54]. However, to the best of our knowledge, literature is very limited regarding the relation between chronotype and sleep hygiene in the athlete population. A possible explanation for this result could be the active pre-sleep routine of evening-type individuals. Being more active in late-afternoon and evening period of the day, evening-types are more likely to engage in active behaviors before sleep, obtaining higher scores in the SHI questionnaire.

The present study had some limitations. First, we evaluated variables using subjective methods (questionnaires) rather than objective measures like actigraphy or polysomnography, which could have provided more validated and reliable data. Additionally, the data collection involved retrospective self-report questionnaires (PSQI and SHI), potentially susceptible to recall bias. In the literature, there are many questionnaires specific to athlete populations (i. e. Athlete Sleep Screening Questionnaire [55], Athlete Sleep Behavior Questionnaire [56], etc.). However, the questionnaires used in the present study are scientifically validated for the assessment of specific area (chronotype [MEQ], quality of sleep [PSQI], and sleep hygiene [SHI]). Second, we did not control the duration and frequency of naps, which would have helped to better understand the total time of sleep during the 24 hours [57], because the questions in the questionnaires completed by the athletes were not detailed enough. Indeed, the literature indicates that athletes participating in individual sports exhibit slightly more frequent naps compared to their counterparts in team sports [7]. Finally, we did not assess whether athletes slept individually or with a partner, and it is well-established that these two different conditions can have an impact on sleep quality and architecture [58]. Additionally, no information about participants' daily routine and occupation (assuming some athletes have second jobs) was obtained. However, to the best of our knowledge, this is the first study that assesses the relationship between chronotype, sleep quality, and sleep hygiene in Italian athletes, providing important information about sleep between sport disciplines and levels of competition. Future research in this area should focus on understanding the sleep habits of Italian athletes using objective methodologies, such as actigraphy. Subsequently, the impact of sleep hygiene strategies on improving sleep quality and performance should be evaluated in athletes of different disciplines and level of competition, due to the different sleep characteristics and habits of samples. This approach will provide valuable insights into how better sleep practices can enhance athletic performance and overall well-being.

In conclusion, the importance of sleep in supporting athletes' recovery is often underestimated. Moreover, the correlation between chronotype, sleep hygiene, and the quality of sleep among athletes remains unclear. Our study provides new insights in this field: a positive correlation was found between PSQI and SHI, and a negative correlation was detected between MEQ and SHI; also, approximately half of sample had poor sleep quality. For these reasons, it is important for coaches to shift their focus towards enhancing performance by also recognizing the profound benefits of adequate recovery. The subjective questionnaires administered in this study could be useful in guiding decision-making regarding sleep interventions in athletes. By capturing athletes' perceptions and experiences related to sleep quality, duration, and patterns, questionnaires and sleep diary could offer valuable insights into the individualized needs of athletes in optimizing their sleep routines. Given the intra- and inter-individual variability [59, 60] and the positive effect of individualized sleep hygiene strategies on athlete's sleep [5], coaches should implement individual sleep education sessions in the daily routine of their athletes.

Funding

This research did not receive any specific grant funding from public agency, commercial, and not-for-profit sectors.

Conflict of Interest

The authors declare that they have no conflict of interest.

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